"Electronic circuits are already at least 10 million times faster than the electrochemical information processing in our interneuronal connections. Machines can share knowledge instantly, whereas we biological humans do not have quick downloading ports," he says.

Thus "Turing Test level" machines will be able to combine human level intelligence with the powerful ways in which machines already excel. In addition, machines will continue to grow exponentially in their capacity and knowledge. It will be a formidable combination.

Recreating the brain

Every conversation about true artificial intelligence will go round and round and wind up at the same point: are we able to fully recreate the human brain? Asimov's vision of the positronic brain might be a fair distance away, but a group of researchers has made rapid strides in recreating the human brain.

At the Ecole Polytechnique Fédérale de Lausanne (EPFL) research institute in Switzerland, scientists are working to create a simulation of the human brain on a supercomputer. The idea of the Blue Brain project is to reverse-engineer how the human brain works.

The first step the scientists took was study the brain of a rat, since rat brains have been shown to be the most biologically similar to humans in the mammalian world. The crux of the project was a virtual revisualisation of the neocortical column – the most evolved part of the brain which is responsible for diverse tasks, the primary among them being reasoning. Although the human brain has a lot more neocortical columns than a rat, the researchers simply wanted to create one column, with the notion that if they can make one, they can make many more.

In 2006, the project tasted success and drew high praise from all quarters. The team is now working on remodelling the entire rat brain in its simulation on a super-computer, after which they will look at doing the same with a cat's brain (more complex), a monkey's brain (even more complex) and then finally a human brain.

But the scientists are already working alongside to turn their virtual representation into a real, physical one. After a lot of analysis and trial-and-error, the team finally decided that carbon nanotubes were the way forward in creating a physical circuit capable of replicating the neurons of the brain. Like neurons, carbon nanotubes are highly electrically conductive, forming extremely tight contacts with neuronal cell membranes. Unlike the metal